# The Clock That Never Ticks

A machine-checked path from the dilogarithm
to quantum speed limits in Lean 4

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**Abstract**

Mathlib's source contains a comment referring to the dilogarithm $\mathrm{Li}_2$ — a function the library does not have. We fill that hole and follow the questions where they lead. The result is a `sorry`-free Lean 4 development containing the real dilogarithm with Euler's reflection identity, Landen's transformation, the duplication formula, and the golden-ratio ladder $\mathrm{Li}_2(1/\varphi^2) = \pi^2/15 - \ln^2\varphi$, $\mathrm{Li}_2(1/\varphi) = \pi^2/10 - \ln^2\varphi$ (derived from a $3\times 3$ linear system — no five-term relation needed); the Rogers value $\mathrm{L}(1/\varphi^2) = \pi^2/15$, equivalently the Lee–Yang effective central charge $c_{\mathrm{eff}} = 2/5$; the Clausen function $\mathrm{Cl}_2$ and *Catalan's constant* $G = \mathrm{Cl}_2(\pi/2)$; the *Fejér–Jackson inequality* $\sum_{k=1}^{M}\sin(k\theta)/k > 0$ on $(0,\pi)$; the positivity $\mathrm{Cl}_2(\theta) \geq \sin(\theta)/2$ by an asymptotics-free Abel summation; and the *Margolus–Levitin* and (an $L^1$ form of the) *Mandelstam–Tamm* quantum speed limits. When version 1 appeared (June 2026) none of these was in Lean; Isabelle's Archive of Formal Proofs already had the polylogarithm of integer order with its duplication formula (Eberl, 2023), which our search had missed. The pieces assemble into the ending the title promises: the quantum state with populations $\propto 1/n^2$ on equally spaced levels has divergent first and second energy moments — both textbook speed limits give no bound — and yet *never* reaches a state orthogonal to itself (a "tick", in this paper, is exactly that). Version 2 is exact about why. Its lowest level carries $6/\pi^2 > 1/2$ of the population, which alone forbids orthogonality (Uffink's cut-off inequality), and the fractional-moment speed limits of Zieliński and Zych reduce, as their order tends to zero, to exactly this criterion. The dilogarithm says more: the modulus of the autocorrelation never drops below $1/2$, its value at half period. For the whole family $p_n \propto n^{-s}$, $s > 1$, we formalize the exact floor $\inf_\theta |S(\theta)| = 1 - 2^{1-s}$, from $|\mathrm{Li}_s(z)| \geq \eta(s)$ on the unit circle for every real $s > 0$; it holds also for $1 < s < s^* \approx 1.7286$, where no level dominates and none of the Margolus–Levitin, Mandelstam–Tamm or Zieliński–Zych bounds certifies that orthogonality never occurs. We also formalize the Zieliński–Zych bound for orders up to 1, including the inequality behind it, and exact examples showing that log-convex or three times monotone populations need not have their smallest autocorrelation modulus at half period. We are exact about the boundary: every identity and inequality here is classical; the contribution is the machine-checked development, its assembly, and the place of the zeta-state theorem among the speed limits. Every named theorem depends only on the three standard axioms.

*Blueprint.* Every result of the paper with its dependencies and its Lean and Isabelle code: https://karlesmarin.github.io/blueprints/dilogarithm-clock/.

*Formalized with AI assistance (Claude, Anthropic); the mathematics and all claims are the author's responsibility. The methodology and the exact boundary of the contribution are discussed in Section 8.

*How to read this paper.* It is written as the story of a chain of questions, Q1 to Q5 — each section opens with its question and closes by handing the reader the next. One more question (*how fast can a quantum state evolve?*) did *not* grow out of the chain: it was born independently, from asking which part of the "universe as a computer" metaphor is an actual theorem, and it *crossed* the chain at its end. Version 2 adds Q6: what decides orthogonality when the easy argument fails. Figure 1 is the honest map, crossing included. If you read only one thing, read Section 6 from Theorem 4 on and Theorem 5; the inventory of the formalizations (Lean and Isabelle) is Appendix B, and what changed in version 2 is Appendix C.

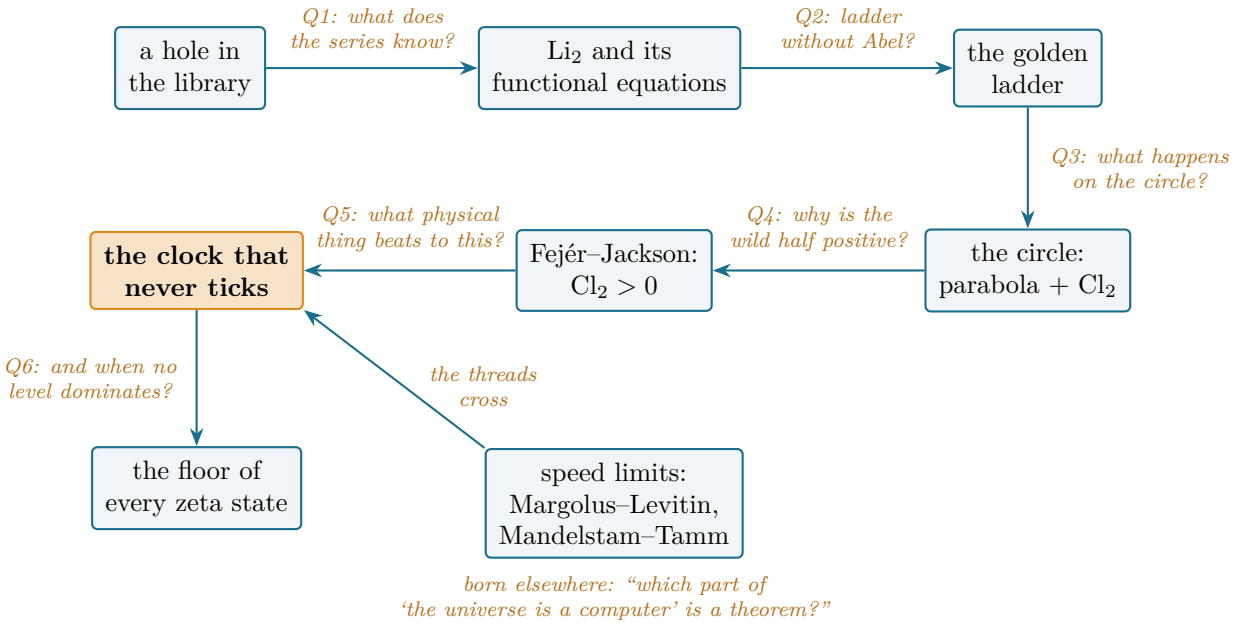



Figure 1: The honest map. The dilogarithm thread runs along the top and back; the quantum thread (bottom) was born independently and *crossed* it at the end — the crossing, not the chain, is where the title theorem lives. Version 2 adds the last arrow. Every node is a block of `sorry`-free Lean.

## 1 The question that starts it: why does the library cite a function it does not have?

Few functions have been discovered as many times as the dilogarithm. Euler wrote down its series in the 1760s; within a century it had been met again by Spence, by Abel in the study of its functional equations, by Lobachevsky in the measurement of hyperbolic space, and by Kummer — each giving it a different name and a different reason to care. Two centuries later it reappeared, unbidden, in the finite parts of Feynman integrals and in the thermodynamics of integrable systems. It is the rare non-elementary function that now and then consents to a closed form, and the catalogue of those closed forms reads like a tour of mathematics. So there is something almost comic in the fact that, in 2026 — after a decade of one of the most ambitious formalization efforts ever attempted — the dilogarithm was not in the library at all. The machines we now use to certify theorems had never been told what Li$_2$ is. They had, however, been told to mention it.

The file `Mathlib/Analysis/SpecialFunctions/Integrals/LogTrigonometric.lean` of Mathlib — the mathematical library of the Lean 4 proof assistant, on which a substantial part of contemporary formalized mathematics is built — describes one of its integrals by referring to the dilogarithm

$$\mathrm{Li}_2(x) \;=\; \sum_{n\geq 1} \frac{x^n}{n^2}.$$

The function itself is absent. When version 1 of this paper appeared (June 2026) it was absent from Mathlib, and the other Lean developments of it that our October search found date from September 2026 on. It was *not* absent from every proof assistant, as version 1 said: Isabelle's Archive of Formal Proofs has had, since November 2023, Eberl's polylogarithm $\mathrm{Li}_s$ of integer order, with its derivative, $\mathrm{Li}_1(z) = -\ln(1-z)$ and the duplication formula [24]. The reflection and Landen identities, the golden values, the Rogers function, the Clausen function, Catalan's constant, the Fejér–Jackson inequality and the quantum speed limits are not there, and we found none of them in the other proof assistants we searched; the existing quantum libraries (CoqQ, SQIR, Isabelle Marries Dirac) formalize circuits, not dynamical bounds.[1]

How does a function this central stay unformalized for a decade of Mathlib? Because it sits in an odd blind spot: too special to be infrastructure, too analytic to be algebra, and guarded — as we will see — by exactly one genuinely delicate limit. The genesis of this project was an audit: a fifty-digit numerical verification of the dilogarithm identities in a recent preprint of Alha [23]. Auditing someone else's dilogarithms sent us to the library for a reference implementation; the library sent us back empty-handed. We decided to fill the hole, and to keep asking the next question until the questions stopped. They stopped five sections later, inside a quantum mechanics problem — and, in version 2, one section after that. This paper is the travelogue.

## 2 Q1: what does the series know about its own derivative?

*History bead.* The function is Euler's (1768); the reflection identity below is his. Landen's transformation is from 1780. Everything in this section is classical (Lewin [1], Zagier [2]); the Lean is `Dilog/Basic.lean`.

A power series differentiates termwise, and the dilogarithm's derivative collapses to elementary functions:

$$\mathrm{Li}_2'(x) \;=\; \sum_{n\geq 1} \frac{x^{n-1}}{n} \;=\; -\frac{\ln(1-x)}{x} \qquad (|x|<1,\; x\neq 0).$$

In Lean this single line is the analytic core of the whole development — term-by-term differentiation under a geometric majorant (`hasDerivAt_tsum_of_isPreconnected`), then the Mercator series to identify the sum. Once the derivative is known, functional equations become mechanical:

[1] Searches of June 2026: Loogle and `grep` over Mathlib (zero `quantum` declarations; no `dilog`/`polylog`/`Clausen`, and Catalan *numbers* only); web searches over the Archive of Formal Proofs and the Rocq/Coq ecosystems, which missed Eberl's entry. Searches of October 2026: the full Archive of Formal Proofs (grep of every theory), and the HOL Light, HOL4, Metamath, Mizar, Rocq/Coq (CoRN, Coquelicot, MathComp-Analysis, Interval), PVS and Lean 3 libraries. Since June, two independent Lean 4 developments have formalized the real dilogarithm with reflection, Landen and the golden values (lean-DLMF, first commit September 2026 [25]) and the dilogarithm with Catalan's constant (LeanPolyLog, September 2026 [26]); a polylogarithm pull request to Mathlib is under review. Absence of evidence, with that caveat.

**Recipe: a functional equation in three moves**

To prove a combination $F$ of dilogarithms and logarithms is constant on an interval: (1) differentiate termwise and check $F' = 0$ by chain/product rules plus `field_simp; ring`; (2) conclude $F$ is constant by mean-value machinery (`constant_of_has_deriv_right_zero` on subintervals); (3) evaluate the constant as a *boundary limit*. Step (3) is where the one delicate limit of the subject lives: $\ln x \cdot \ln(1-x) \to 0$ at the endpoint, tamed as $\big(\ln(1-x)/x\big) \cdot \big(x \ln x\big) \to (-1) \cdot 0$ via the slope characterization of the derivative and `tendsto_log_mul_rpow_nhdsGT_zero`. Continuity of $\mathrm{Li}_2$ up to the boundary is the Weierstrass $M$-test (`continuousOn_tsum`).

**Theorem 1** (`Li2_add_Li2_one_sub`, `Li2_landen`, `Li2_sq`, `Li2_one_half`)**.** *On the stated domains,*

$$\begin{aligned} \mathrm{Li}_2(x) + \mathrm{Li}_2(1-x) &= \tfrac{\pi^2}{6} - \ln x \, \ln(1-x) && x \in (0,1) \\ \mathrm{Li}_2(x) + \mathrm{Li}_2\Big(\tfrac{x}{x-1}\Big) &= -\tfrac{1}{2}\ln^2(1-x) && x \in (0, \tfrac{1}{2}) \\ \mathrm{Li}_2(x^2) &= 2\big(\mathrm{Li}_2(x) + \mathrm{Li}_2(-x)\big) && |x| \le 1 \\ \mathrm{Li}_2(\tfrac{1}{2}) &= \tfrac{\pi^2}{12} - \tfrac{\ln^2 2}{2}. \end{aligned}$$

Two honesty notes, both visible in the Lean statements. The Landen domain stops at $x = \frac{1}{2}$: past it, the argument $x/(x-1)$ leaves the disc of convergence, and a power-series definition does not analytically continue — we do not pretend it does. And the duplication formula needs no calculus at all: it is a pure rearrangement of the series by even and odd indices (`tsum_even_add_odd`), the formal mirror of how Euler would have proved it. (Isabelle's AFP proves it too, for every integer order [24].)

*From theory to practice.* The dilogarithm is the standard weight-2 function of one-loop Feynman integrals and of hyperbolic and $K$-theoretic invariants; textbooks reach for it constantly. The practical value of formalizing it is modest and specific: a verified `Li2` with its functional equations — including the one delicate boundary limit — that downstream Lean developments can import rather than re-derive.

*The turn.* Three functional equations in hand, the classical showpieces of the subject are the golden-ratio evaluations — and one route to them goes through Abel's five-term relation, a two-variable identity this paper does *not* formalize (its sequel does [27]). Are we stuck?

## 3 Q2: can the golden ladder be climbed without Abel's relation?

*History bead.* The values are Landen's (1780); the two-term sum is from Rogers' 1907 memoir [4]. Their role as a central charge belongs to the thermodynamic Bethe ansatz of the 1990s [15, 16, 17].

The answer is yes, and the key is two arithmetic accidents of the golden ratio $\varphi = (1+\sqrt{5})/2$, both one-liners from $\varphi^2 = \varphi + 1$:

$$\frac{1}{\varphi} + \frac{1}{\varphi^2} = 1 \qquad \text{and} \qquad \frac{1/\varphi^2}{1/\varphi^2 - 1} = -\frac{1}{\varphi}.$$

The first says $1/\varphi$ and $1/\varphi^2$ are a *reflection pair*; the second says *Landen maps* $1/\varphi^2$ *to* $-1/\varphi$ — and note $1/\varphi^2 \approx 0.382 < \frac{1}{2}$, inside our proven Landen domain. Reflection at $1/\varphi^2$, Landen at $1/\varphi^2$, duplication at $1/\varphi$: three linear equations, three unknowns, and `linarith` closes the system.

$$\mathrm{Li}_2\Big(\tfrac{1}{\varphi^2}\Big) = \tfrac{\pi^2}{15} - \ln^2\varphi, \qquad \mathrm{Li}_2\Big(\tfrac{1}{\varphi}\Big) = \tfrac{\pi^2}{10} - \ln^2\varphi, \qquad \mathrm{Li}_2\Big(-\tfrac{1}{\varphi}\Big) = \tfrac{\ln^2\varphi}{2} - \tfrac{\pi^2}{15}.$$

No five-term relation is needed — not a new route (the values are Landen's, older than Abel's relation), but one the formalization makes explicit. In the Rogers normalization $\mathrm{L}(x) = \mathrm{Li}_2(x) + \frac{1}{2}\ln x \ln(1-x)$ the $\ln^2$ defects cancel (`rogersL_inv_goldenRatio_sq`, `rogersL_gold_sum`):

$$\mathrm{L}\Big(\tfrac{1}{\varphi^2}\Big) = \frac{\pi^2}{15}, \qquad \mathrm{L}\Big(\tfrac{1}{\varphi}\Big) + \mathrm{L}\Big(\tfrac{1}{\varphi^2}\Big) = \frac{\pi^2}{6} = \mathrm{L}(1),$$

and the first, normalized, is the effective central charge of the Lee–Yang thermodynamic Bethe ansatz, $c_{\text{eff}} = \mathrm{L}(1/\varphi^2)/\,\mathrm{L}(1) = 2/5$ — to our knowledge the first TBA dilogarithm identity verified in a proof assistant.

*From theory to practice.* The value $c_{\text{eff}} = 2/5$ is the effective central charge of the Lee–Yang model, a non-unitary minimal CFT whose Yang–Lee zeros have been observed experimentally in spin systems. The identity formalized here is the simplest entry of the thermodynamic-Bethe-ansatz dilogarithm table; having one machine-checked anchors any later formalization of integrable-model thermodynamics.

*The turn.* The ladder spends the function's life *inside* the interval of convergence. The natural next territory is the boundary itself. What does the series do on the unit circle?

## 4 Q3: what happens on the circle?

*History bead.* Clausen introduced his function in 1832 [3]; Catalan's constant is from his 1865 memoir, and its irrationality is, famously, still open. The cosine series below is the Fourier expansion of the Bernoulli polynomial $B_2$, which Mathlib already owns (`hasSum_one_div_nat_pow_mul_cos`); the Lean of this section is `Dilog/Clausen.lean`.

On the circle the dilogarithm splits into a tame half and a wild half:

$$\mathrm{Li}_2(e^{i\theta}) \;=\; \underbrace{\sum_{n\ge1}\frac{\cos n\theta}{n^2}}_{=\,\frac{\pi^2}{6}-\frac{\pi\theta}{2}+\frac{\theta^2}{4}} \;+\, i\,\underbrace{\sum_{n\ge1}\frac{\sin n\theta}{n^2}}_{=\,\mathrm{Cl}_2(\theta)} \qquad (\theta\in[0,2\pi]).$$

The real part is an *elementary parabola* (`hasSum_cos_div_sq`, from Mathlib's Bernoulli–Fourier machinery at $k=1$). The imaginary part is not elementary at all: it is the **Clausen function**, a genuinely new special function — the reason this section exists. At the quarter circle it produces a celebrity:

$$G \;=\; \mathrm{Cl}_2\Big(\frac{\pi}{2}\Big) \;=\; \sum_{k\ge0}\frac{(-1)^k}{(2k+1)^2} \;=\; 0.9159\ldots,$$

**Catalan's constant**, absent from every proof assistant we searched when version 1 appeared (it has since been defined independently in LeanPolyLog [26]) (`catalanConst`, `Cl₂_pi_div_two`; the proof is the same even/odd rearrangement as the duplication formula — even-indexed sines vanish, odd-indexed give the alternating series). Basic symmetries come for free: $\mathrm{Cl}_2(0) = \mathrm{Cl}_2(\pi) = 0$, $\mathrm{Cl}_2$ is odd, and $\mathrm{Cl}_2(2\pi-\theta) = -\mathrm{Cl}_2(\theta)$.

*From theory to practice.* $\mathrm{Cl}_2$ is, up to scale, Lobachevsky's function: the volume of an ideal hyperbolic tetrahedron is a sum of three $\mathrm{Cl}_2$ values, so this is the analytic core of hyperbolic 3-manifold and knot-complement volumes. Catalan's constant appears as the dimer entropy $G/\pi$ of the square lattice. A verified $\mathrm{Cl}_2$ is a concrete step toward machine-checked hyperbolic volumes and lattice-entropy constants.

*The turn.* Plot the wild half (Figure 2) and one suspicion dominates everything: between 0 and $\pi$ the function looks *positive*. Two values and a hunch are not a theorem. Why is it positive? That question is older than it looks: it is Fejér's, from 1910.

## 5 Q4: why is the wild half positive?

*History bead.* Fejér raised and partially proved the positivity of the sine partial sums in 1910; Jackson's proof is from 1911 [5, 6]. It is the grandfather of the theory of positive trigonometric sums (Vietoris, Askey). The Lean is `Dilog/FejerJackson.lean`.

**Theorem 2** (`fejerSum_pos`). *For every $M \geq 1$ and $\theta \in (0, \pi)$:* $\displaystyle\sum_{k=1}^{M} \frac{\sin k\theta}{k} > 0.$

*Hand-checkable case.* $M = 2$: $\sin\theta + \frac{1}{2}\sin 2\theta = \sin\theta\,(1 + \cos\theta) > 0$ on $(0, \pi)$. The point of the formalization is that the same certainty now covers every $M$.

The proof is the classical interior-minimum induction, with two touches worth recording. First, the Dirichlet kernel is handled *multiplicatively* — no division, no removable singularity:

$$2\sin\tfrac{\theta}{2}\sum_{k=1}^{M}\cos k\theta = \sin\big((M+\tfrac{1}{2})\theta\big) - \sin\tfrac{\theta}{2},$$

a telescope of $\sin((k+\frac{1}{2})\theta)$ via the product-to-sum identity (`Finset.sum_range_sub`). Second, at an interior local minimum the critical equation $\sin((M+\frac{1}{2})\theta_0) = \sin(\theta_0/2)$ *factors* (`Real.sin_sub_sin`) as $\sin(M\theta_0/2)\cos((M{+}1)\theta_0/2) = 0$, and its two branches force

$$\sin(M\theta_0) = 0 \qquad \text{or} \qquad \sin(M\theta_0) = \sin\theta_0 > 0\,:$$

either way the last term of the sum is nonnegative at the minimum, so the minimum value is at least the previous partial sum, $S_M(\theta_0) \geq S_{M-1}(\theta_0) > 0$ by induction. Boundary values vanish; compactness closes.

From Fejér–Jackson to the Clausen function is an Abel summation, and here the formalization found a small simplification: *no asymptotics are needed.* Writing the $(N{+}1)$-term partial sum of the $\mathrm{Cl}_2$ series in terms of the Fejér–Jackson sums $S_M$,

$$\sum_{n=1}^{N+1}\frac{\sin n\theta}{n^2} = \frac{S_{N+1}(\theta)}{N+1} + \sum_{k=1}^{N} S_k(\theta)\Big(\frac{1}{k} - \frac{1}{k+1}\Big),$$

every piece on the right is positive, and the $k = 1$ summand alone equals $\sin(\theta)/2$. Every partial sum is therefore $\geq \sin(\theta)/2$ — no harmonic-number bounds, no tail estimates — and the limit inherits the bound (`Cl₂_pos`; as a theorem of its own, `sin_div_two_le_Cl₂` in [28]):

$$\boxed{\mathrm{Cl}_2(\theta) \;\geq\; \frac{\sin\theta}{2} \;>\; 0 \qquad (0 < \theta < \pi).}$$

*From theory to practice.* The Fejér kernel underlies Cesàro summability of Fourier series — the standard remedy for Gibbs oscillation in signal reconstruction and filter design — and positivity of trigonometric partial sums is a recurring gate in geometric function theory. The verified inequality is a reusable building block for formalized approximation theory.

*The turn — the big one, and the honest one.* Step back and look at what we are holding: a sum $\sum p_n e^{-in\theta}$ with positive weights $p_n \propto 1/n^2$, whose imaginary part never vanishes on $(0, \pi)$. A physicist reads that expression instantly — it is the *autocorrelation of a quantum state*, the inner product between a state and its own time evolution, for populations $p_n$ on equally spaced energy levels. We must be honest about how this reading entered the project, because it did *not* grow out

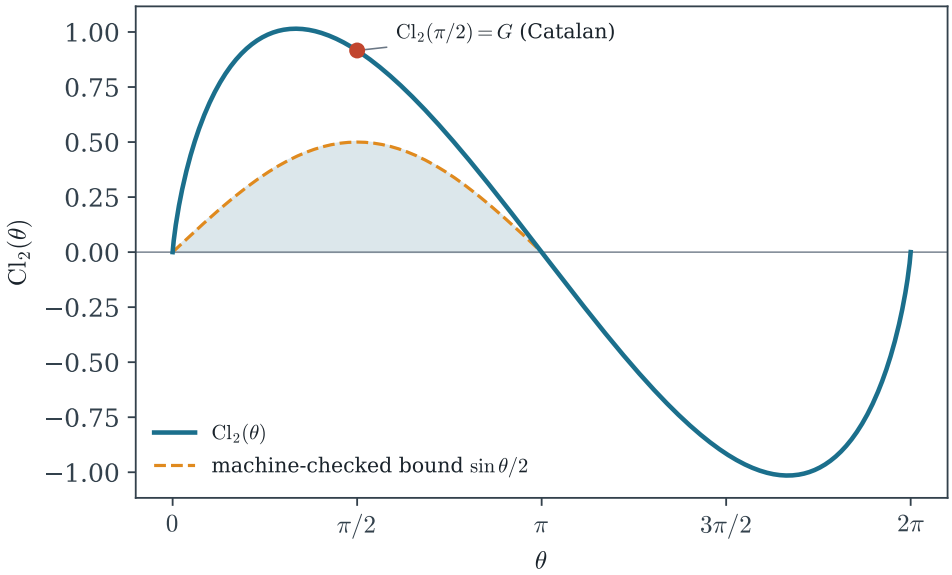



Figure 2: The Clausen function on $[0, 2\pi]$ with the machine-checked lower bound $\sin(\theta)/2$ (shaded) on $(0, \pi)$, and Catalan's constant at the quarter circle. The generating script asserts the bound on a 400-point grid at 30 digits.

of the chain of questions: a parallel thread — born from asking which part of the old "universe as a computer" metaphor is an actual theorem — had already formalized the two classical bounds on how fast any quantum state can evolve. At this point the two threads crossed: the circle had been measuring *time* all along, and the positivity theorem we had just proved was a statement about a quantum system that cannot reach an orthogonal state. What do the speed limits say about it?

## 6 Q5: what physical thing beats to this function?

*History bead.* Mandelstam and Tamm bounded the orthogonalization time by the energy *spread* in 1945 [8]; Margolus and Levitin bounded it by the mean energy in 1998 [9]. Together they are the textbook "quantum speed limits": how many distinguishable states can a system visit per unit time, on a given energy budget. The Lean is `QSL/`, and, for what version 2 adds, `Physics/QuantumMechanics/SpeedLimit/` in the formalizations repository [28].

A state $|\psi_0\rangle = \sum_n c_n |E_n\rangle$ evolves with autocorrelation $S(t) = \sum_n p_n e^{-iE_n t}$, $p_n = |c_n|^2$, and reaches an orthogonal state at time $\tau$ iff $S(\tau) = 0$. This is a statement about a scalar function of a discrete nonnegative spectral measure, and that is how we formalize it — the Hilbert space is bookkeeping. Orthogonality enters as two real conditions ($\mathrm{Re}\, S = \mathrm{Im}\, S = 0$); Margolus–Levitin needs no complex numbers at all.

**Theorem 3** (`margolus_levitin`, `mandelstam_tamm_L1`; $\hbar = 1$)**.** *Let $p_n \geq 0$ with $\sum_n p_n = 1$ be the populations of levels $E_n \geq 0$, with finite mean $\langle E \rangle = \sum_n p_n E_n$ and finite $D_1$ about a real centre $\bar{E}$. If $S(\tau) = 0$ with $\tau \geq 0$, then*

$$\pi \;\leq\; 2\,\langle E \rangle\,\tau \qquad \textit{and} \qquad 1 \;\leq\; D_1\,\tau, \quad D_1 = \textstyle\sum_n p_n |E_n - \bar{E}|.$$

The heart of Margolus–Levitin is one inequality of independent interest, $\cos x \geq 1 - \frac{2}{\pi}(x + \sin x)$ for $x \geq 0$ (equality at 0 *and* $\pi$), which we prove by a rise-then-fall argument: both endpoints zero,

split at $x_0 = 2\arctan(2/\pi)$. The same shape — both endpoints at zero, and the critical-point equation analysed at an interior extremum — carries the Fejér–Jackson proof above; that is a transferable technique. Since $D_1 \le \Delta E$ by Cauchy–Schwarz, the second bound implies the textbook weak Mandelstam–Tamm $\tau \ge 1/\Delta E$ (`mandelstam_tamm_weak`); the sharp constant $\pi/2$ needs the geodesic argument and is left as a road (§9).

Now the dénouement. Take populations $p_n = \frac{6}{\pi^2 n^2}$ on levels $E_n = n$: the *weight-2 zeta state*. Its autocorrelation *is* our protagonist,

$$S(\theta) = \frac{6}{\pi^2}\,\mathrm{Li}_2(e^{-i\theta}) = \frac{6}{\pi^2}\Big( \underbrace{\tfrac{\pi^2}{6} - \tfrac{\pi\theta}{2} + \tfrac{\theta^2}{4}}_{\text{the parabola of §4}} \; - i \quad \underbrace{\mathrm{Cl}_2(\theta)}_{\text{positive, by §5}} \Big).$$

Its first energy moment $\sum_n p_n\, n$ *diverges*, and so does its second. Theorem 3 does not apply — its moments are not finite — and the two textbook speed limits give no positive bound. (The state is normalized and its evolution is well defined; $S$ is continuous with $S(0) = 1$, so nothing happens instantly either.) In the language of the title, a *tick* is reaching a state exactly orthogonal to the initial one. This state never ticks:

**Theorem 4** (`zetaState_never_orthogonal`)**.** *For every $\theta \in (0, 2\pi)$, the real and imaginary parts of the weight-2 zeta-state autocorrelation are not both zero: the state* never *reaches an orthogonal state.*

*Proof (as in Lean).* On $(0, \pi)$ the imaginary part is $-\frac{6}{\pi^2}\mathrm{Cl}_2(\theta) \ne 0$ by the Fejér–Jackson bound. At $\theta = \pi$ the real part is $\frac{6}{\pi^2}$ times the parabola value $-\pi^2/12$ (which is $-\eta(2)$), that is $-\frac{1}{2} \ne 0$. On $(\pi, 2\pi)$, the reflection $\mathrm{Cl}_2(2\pi - \theta) = -\mathrm{Cl}_2(\theta)$ hands the sign back to the first case. □

*Why, in one line.* Version 1 told this as a surprise; it is not one. The lowest level carries $p_1 = 6/\pi^2 \approx 0.608$ of the population, more than half, and a level that carries more than half of the population cannot be cancelled by all the others:

$$|S(t)| \;\ge\; p_1 - \sum_{n \ge 2} p_n \;=\; 2p_1 - 1 \;\approx\; 0.216 \qquad \text{for every } t.$$

This is Uffink's cut-off inequality $|A(t)| \ge \alpha|\bar{A}(t)| - (1 - \alpha)$ in the form of Andrews [12, Eq. (21)] (attributed there to [10]), with the cut-off isolating one level, so that $|\bar{A}| \equiv 1$ ($\bar{A}(t) = e^{-iE_1 t}$) and $\alpha = p_1$. In Lean, for any probability distribution and any phases of modulus one, it is `two_mul_sub_one_le_norm_tsum`. So Theorem 4 needs neither the Clausen function nor Fejér–Jackson; what it does need is a bound that does not go through the first two energy moments, which is exactly where the textbook limits fail. (A dominant level forbids exact orthogonality, not fast evolution: the state $\sqrt{0.6}\,|0\rangle + \sqrt{0.4}\,|\Omega\rangle$ never ticks, but reaches its smallest overlap, 0.2, at $t = \pi/\Omega$, as early as one likes.)

*What the moments can see.* Margolus–Levitin and Mandelstam–Tamm use the first two moments, which are infinite here. Zieliński and Zych [11] gave a whole family that uses the fractional moments $M_\alpha = \langle (E - E_0)^\alpha \rangle$, $\alpha > 0$, which are finite for the zeta state when $\alpha < 1$:

$$\tau \;\ge\; \frac{\pi}{(2M_\alpha)^{1/\alpha}}.$$

Their proof integrates the inequality $x^\alpha - \pi^\alpha/2 + (\pi^\alpha/2)\cos x + \alpha\pi^{\alpha-1}\sin x \ge 0$ ($x \ge 0$), which they state without proof; we prove it and the bound in Lean for $0 < \alpha \le 1$ (`zz_inequality`, `zielinski_zych`), the range Proposition 1 below needs; the case $\alpha > 1$ is not formalized. Their argument only needs $E \ge E_0$ on the energies that carry population, so $E_0$ may be taken to be the lowest populated level.

**Proposition 1** (`tendsto_fractionalMoment`, `zz_bound_tendsto_atTop`, `zz_bound_tendsto_zero`)**.** *Let $E_0$ be the lowest populated level, $p_0$ its population, the other populated levels strictly above it, and some fractional moment finite. As $\alpha \to 0^+$, $M_\alpha \to 1 - p_0$, so the Zieliński–Zych bound tends to $+\infty$ if $p_0 > 1/2$ (and $p_0 < 1$) and to 0 if $p_0 < 1/2$.*

*Proof.* For $\alpha \to 0^+$ each term $p_n(E_n - E_0)^\alpha$ tends to $p_n$ if $E_n > E_0$ and is 0 for $E_n = E_0$; dominated convergence gives $M_\alpha \to 1 - p_0$. Then $(2M_\alpha)^{1/\alpha} = \exp\left(\frac{1}{\alpha}\ln(2M_\alpha)\right)$ tends to 0 when $2(1-p_0) < 1$ and to $+\infty$ when $2(1-p_0) > 1$. □

So the whole family certifies $\tau_\perp = \infty$ for the weight-2 zeta state — and it does so through the triangle-inequality criterion applied to one level only, "the lowest level holds more than half" (a dominant level higher up, as in $p = (0.2, 0.8)$, is invisible to the moments measured from below; when the spectrum is bounded above, the same argument applied to $E_{\max} - H$, which leaves $|S|$ unchanged, sees it). At the threshold $p_0 = \frac{1}{2}$ a short expansion, $M_\alpha = \frac{1}{2} + \alpha L + o(\alpha)$ with $L = \sum_{n>0} p_n \ln(E_n - E_0)$ when that sum is finite, gives the limit $\pi e^{-2L}$: for the zeta state at $s = s^*$ (Section 7), 0.7257, consistent with the values computed at small $\alpha$; we have not formalized this case. With the convention of Theorem 3 ($E_0 = 0$, an empty level below the spectrum) the family never certifies it: every term then tends to $p_n$, $M_\alpha \to 1$, and the bound tends to 0. We have not found Proposition 1 stated; it is an easy observation, and we record it because it says what moment-based speed limits can and cannot see. (The truncated weight-2 zeta state appears as a test case for speed limits in [13, Table I, case (c)], where the bounds reported are finite lower bounds; that the state never becomes orthogonal is not discussed there.)

*The floor.* The dilogarithm says more than non-vanishing. On the whole circle $|S| \geq \frac{6}{\pi^2} \cdot \frac{\pi^2}{12} = \frac{1}{2}$, with equality at $\theta = \pi$ (`zetaState_norm_ge_half`; version 1 had this only numerically). The bound $|\mathrm{Li}_2(z)| \geq \pi^2/12$ on $|z| = 1$ follows from a theorem of Fejér on four times monotone coefficients [7]; for $\mathrm{Li}_2$ it is stated by Boyer and Parry [20, Appendix A, Lemma 8]. So the survival probability $|S|^2$ never drops below $\frac{1}{4}$ (Figure 3). A clock with an unbounded energy budget whose hand never completes a tick — not because the budget is spent on a heavy spectral tail, as version 1 put it (the tail only makes the moments infinite), but because more than half of the state sits on one level. Energy is necessary for reaching an orthogonal state quickly (Margolus–Levitin); a dominant level rules out reaching one at all, whatever the energy.

*From theory to practice.* Quantum speed limits set the minimum time for a gate, a measurement, or a state transfer on a given energy budget; they recur across quantum computing, metrology, and quantum thermodynamics (battery charging, optimal control). The zeta state carries a practical lesson: a large energy budget does not buy an orthogonal state if more than half of the weight stays on one level — reaching an orthogonal state needs the weight spread over levels, not merely energy in quantity.

*The turn.* If the easy argument is "one level holds more than half", the honest question is what happens when no level does.

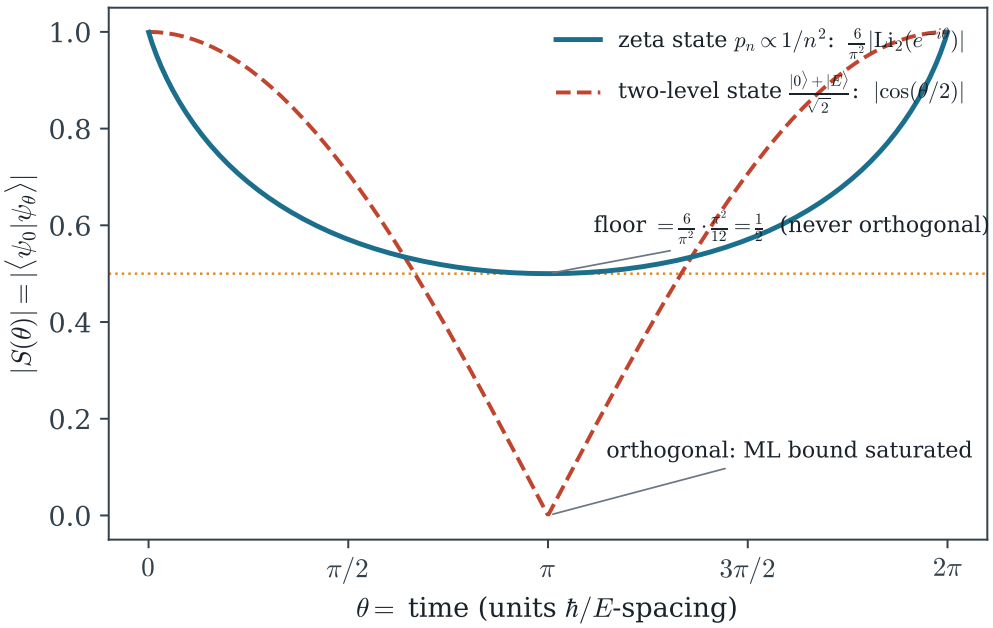



Figure 3: Two states with the same level spacing. The two-level state $(|0\rangle + |E\rangle)/\sqrt{2}$ (dashed) saturates Margolus–Levitin and strikes orthogonality at $\theta = \pi$. The zeta state (solid), with *infinite* mean energy, has $|S| \geq \frac{1}{2}$ (a theorem since version 2: `zetaState_norm_ge_half`): the clock that never ticks. The script asserts the floor equals $\eta(2) = \pi^2/12$ at $\theta = \pi$.

## 7 Q6: what decides orthogonality when no level dominates?

Take the whole family of zeta states, $p_n = n^{-s}/\zeta(s)$ on the levels $E_n = n$, $s > 1$. The first energy moment diverges for $s \leq 2$ and the second for $s \leq 3$. The lowest level carries $p_1 = 1/\zeta(s)$, and $p_1 < \frac{1}{2}$ exactly when $\zeta(s) > 2$, that is for $1 < s < s^*$ with $s^* = 1.72864723899818\ldots$ ($\zeta(s^*) = 2$). In that range the triangle inequality says nothing, Margolus–Levitin and Mandelstam–Tamm say nothing, and every Zieliński–Zych bound is finite, as is their supremum over $\alpha$ (with $E_0 = 1$, the lowest level: numerically 0.0016 at $s = 1.3$, 0.066 at $s = 1.5$, 0.45 at $s = 1.7$; the bound tends to 0 at both ends of the range of $\alpha$, by Proposition 1 and because $M_\alpha \to \infty$ as $\alpha \to s - 1$). These bounds certify a positive time before which the state cannot be orthogonal; none of them decides whether it ever is.

Uffink's cut-off with more than one level does decide it, one state at a time: keeping the first $N$ levels, $|S| \geq \min_\theta \left|\sum_{n\leq N} p_n e^{-in\theta}\right| - \sum_{n>N} p_n$, and the right-hand side is positive for $N = 3$ at $s = 1.7$, $N = 8$ at $s = 1.5$ and $N = 200$ at $s = 1.3$ (numerically: 0.091, 0.023, 0.015). It needs a certified minimum of a trigonometric polynomial for each $s$, a number of levels that grows as $s \to 1$, and it gives a floor far below the true one (0.384, 0.293, 0.188). Two arguments settle the whole family at once, the second with the exact floor.

The first is the shape of the populations: they decrease strictly, and a convergent power series with strictly decreasing positive coefficients has no zero on the unit circle. For polynomials this, and the description of the zeros on the circle when the decrease is not strict (they are roots of unity, and the coefficients come in groups of equal consecutive values), is Hurwitz's 1913 complement [19, §2] to the Eneström–Kakeya theorem [18]. Version 1 stated it as Remark 1 and wrote that it had not found it in the literature; it is Hurwitz's. In Lean, for summable non-increasing series, it is

`enestrom_kakeya_boundary_series` and `strictAnti_series_ne_zero`. Physically: a state with strictly decreasing populations on equally spaced levels never becomes orthogonal; a state with non-increasing populations that does orthogonalize has its decrements on an arithmetic progression, like the Margolus–Levitin-sharp state $(|0\rangle + |E\rangle)/\sqrt{2}$.

The second gives the exact floor.

**Theorem 5** (`polylog_circle_floor`, `polylog_neg_one`)**.** *For every real $s > 0$ and every $z \neq 1$ on the unit circle, the series $\mathrm{Li}_s(z) = \sum_{n\geq 1} z^n/n^s$ converges and*

$$|\mathrm{Li}_s(z)| \;\geq\; \eta(s) = \sum_{n\geq 1} \frac{(-1)^{n-1}}{n^s},$$

*with equality at $z = -1$.*

**Corollary 1** (`zeta_floor`, `zetaState_floor`; the equality case from `polylog_neg_one` and `tendsto_partialEta_of_one_lt`)**.** *For every $s > 1$ the zeta state never becomes orthogonal, and*

$$|S(\theta)| \;\geq\; \frac{\eta(s)}{\zeta(s)} \;=\; 1 - 2^{1-s} \qquad \text{for every } \theta,$$

*with equality at $\theta = \pi$. For $s = 2$ the floor is $\frac{1}{2}$.*

*Proof of Theorem 5.* The coefficients are moments: $(k+1)^{-s} = \int_0^1 t^k \, d\mu_s(t)$ with the positive measure $d\mu_s = (-\ln t)^{s-1} dt/\Gamma(s)$, so that $\mathrm{Li}_s(z)/z = \int_0^1 d\mu_s(t)/(1-tz)$ inside the disc — the representation of $\mathrm{Li}_s(z)/z$ as a Stieltjes transform of Roth, Ruscheweyh and Salinas [21, p. 3]. For $t \in [0,1)$ and $z = e^{i\theta}$,

$$\mathrm{Re}\,\frac{1}{1-tz} - \frac{1}{1+t} = \frac{t\,(1+\cos\theta)(1-t)}{(1+t)\,|1-tz|^2} \;\geq\; 0.$$

With $F(w) = \mathrm{Li}_s(w)/w$ and $0 < r < 1$, integrating gives $|F(rz)| \geq \mathrm{Re}\,F(rz) \geq \int d\mu_s/(1+rt)$; Abel's theorem and dominated convergence take $r \to 1^-$, $\int d\mu_s/(1+t) = \sum(-1)^k(k+1)^{-s} = \eta(s)$, and $|\mathrm{Li}_s(z)| = |F(z)|$ on the circle. The corollary is the classical identity $\eta(s) = (1-2^{1-s})\zeta(s)$ (`tendsto_partialEta_of_one_lt`). □

The theorem is classical: it is a consequence of Fejér's theorem on four times monotone coefficients, which make $\mathrm{Re}\,f(e^{i\theta})$ decrease on $[0,\pi]$ (partial sums of the transform, [7, p. 104]) and $|f(e^{i\theta})|^2$ decrease on $[0,\pi]$ [7, p. 107]; for $s = 2$ it is Boyer and Parry's lemma [20]. The proof above is the one we formalized; its only estimate is the displayed identity. The same proof gives the floor for every population that is a *mixture of geometric populations*: $p_n = \int t^n \, d\mu(t)$, $n \geq 0$, with $\mu$ a positive measure on $[0,1)$ normalized by $\sum_n p_n = \int d\mu/(1-t) = 1$ (absence of mass at $t = 1$ alone does not make the populations summable: the uniform $\mu$ gives $p_n = 1/(n+1)$) (`le_norm_of_tendsto`; for finite mixtures `antipode_floor`, already in June). Physically these are the populations of mixtures of thermal states of a harmonic oscillator, read as the populations of a pure state on the levels $0, 1, 2, \ldots$: every such state never becomes orthogonal, and the modulus of its autocorrelation is smallest at half period, where it equals $\sum_n (-1)^n p_n$. (A shift of the energy origin, as from levels $n \geq 1$ to $n \geq 0$, multiplies $S$ by a phase: the modulus, which is what orthogonality sees, does not change; the real part and the sign at $\theta = \pi$ do.) We have not found this reading in the physics literature; the closest work on the zeros of survival amplitudes argues from the dominant terms and leaves out equally spaced spectra [14].

*How far the shape condition reaches.* Complete monotonicity is more than the floor needs: Fejér's four times monotone sequences suffice [7, p. 107]. Weaker shape conditions do not, and the

smallest modulus then leaves half period. Both witnesses below are at $z^* = (-7 + i\sqrt{15})/8$, on the unit circle, and compare moduli, not real parts:

| populations $q_n$ (up to normalization) | smallest $\lvert f\rvert$ at half period? | witness (exact), $f = \sum q_n z^n$ |
|---|---|---|
| completely monotone (Hausdorff) | yes | Theorem 5 |
| four times monotone | yes | Fejér [7, p. 107] |
| three times monotone | no | $f = 6 + 3z + z^2 + \frac{1}{1-z/2}$: $\lvert f(z^*)\rvert^2 = f(-1)^2 - \frac{635}{2448}$, $f(-1) = \frac{14}{3}$ |
| log-convex (hence convex) | no | $q_n = (\frac{3}{4})^n$ for $n \le 3$, then ratio $\frac{7}{8}$: $\lvert f(z^*)\rvert^2 = f(-1)^2 - \frac{57}{42200}$, $f(-1) = \frac{47}{80}$ |

Both examples are Lean theorems (`cex3_monotone_three`, `cex3_normSq`, `cexLC_logConvex`, `cexLC_convex`, `cexLC_normSq`), and both were checked exactly in two computer algebra systems. The real parts drop too (`cex3_re`: $\frac{73}{16} < \frac{14}{3}$ at $\cos\theta = -\frac{3}{4}$; `cexLC_re`: $-\frac{381}{27040}$ at $e^{2\pi i/3}$), but a lower real part alone says nothing about the modulus: at $\cos\theta = -\frac{3}{4}$ the modulus of the first example is in fact larger than at $z = -1$ (by $\frac{109}{288}$ in the square). Strictly decreasing populations still never orthogonalize; what fails is that half period is the worst moment. We have not found these examples in the literature.

## 8 The boundary of the contribution

Everything here that is mathematics is classical: Euler (1768), Landen (1780), Clausen (1832), Rogers (1907), Fejér (1910, 1937), Jackson (1911), Hurwitz (1913), Mandelstam–Tamm (1945), Margolus–Levitin (1998), Uffink (1993); the TBA reading of the golden values is from the 1990s; the floor of $\lvert\mathrm{Li}_2\rvert$ on the circle follows from Fejér (1937) and is stated by Boyer and Parry (2021); the fractional-moment bounds are Zieliński and Zych's (2006). We claim no new identity and no new inequality of substance. The contributions are: (i) the machine-checked development itself — in June 2026 the first formalization, as far as we could determine, of each item of the abstract except the polylogarithm and its duplication formula, which Isabelle's AFP had [24]; (ii) three proof-engineering choices that may be reusable: the division-free Dirichlet telescope, the asymptotics-free Abel bound $\mathrm{Cl}_2 \ge \sin(\theta)/2$, and the statements of the speed limits as facts about a scalar spectral measure (fully real for Margolus–Levitin; Mandelstam–Tamm uses the complex autocorrelation); (iii) the assembly of Theorem 4, and, in version 2, its honest place: it follows from one dominant level, which the moment-based speed limits see only in the limit of Proposition 1; (iv) the floor of every normalizable zeta state, $s > 1$ (Corollary 1), as a statement about speed limits where the moment-based bounds are silent, from the analytic floor of Theorem 5, formalized for every real $s > 0$, with the proof of the Zieliński–Zych inequality for $0 < \alpha \le 1$ and the exact examples of the last table. The physical readings in (iii) and (iv) are ones we have not seen stated; we would not be surprised to find them folklore among polylogarithm or speed-limit specialists.

*Related formalization.* We build directly on Mathlib's analytic infrastructure: term-by-term differentiation, the Weierstrass $M$-test, mean-value machinery, and the Bernoulli–Fourier expansion that hands us the parabola for free. Special functions in the proof-assistant literature have centred on $\Gamma$, $\zeta$ and the elementary transcendental functions; Isabelle has the polylogarithm of integer order [24], and since version 1 two Lean 4 developments have formalized parts of the real dilogarithm [25, 26]. On the quantum side, the established libraries (CoqQ, SQIR, Isabelle Marries Dirac) formalize

circuits and gate semantics rather than dynamical bounds on continuous evolution. This development sits in the gap between those bodies of work.

On methodology: the formalization was carried out in collaboration with an AI assistant (Claude, Anthropic; version 2 with Claude Opus), with every step gated by the Lean kernel, every identity checked independently at 20–50 digits *before* formalization, and every named theorem verified by `#print axioms` to depend only on `propext`, `Classical.choice`, `Quot.sound`. Correctness does not rest on the assistant: the Lean kernel checks every proof, so the AI affected the path to each theorem, not its validity. The literature checks of version 2 were also done with AI assistance; every source cited for a statement was read in the original, except Uffink's note, which we cite through Andrews. The mathematics, the mistakes, and the claims remain the author's responsibility.

## 9 Open roads

1. **After the five-term relation.** Abel's five-term relation, the capstone of weight 2, was formalized in the sequel to version 1 [27]; with it, every two-variable dilogarithm identity is within reach of the same machinery.

2. **The complex dilogarithm** and the Bloch–Wigner function; with them, the 2D dimer entropy ($G/\pi$ — Catalan again) and hyperbolic volumes come into reach. (The boundary values $\mathrm{Li}_2(e^{i\theta})$ and $\mathrm{Li}_2(i) = -\pi^2/48 + iG$ are now in Lean [28].)

3. **Sharp Mandelstam–Tamm** ($\pi/2\Delta E$, the geodesic argument) and the arbitrary-fidelity Margolus–Levitin bound — conjectured in 2003, proven only in 2023 [22]: a fresh theorem to formalize.

4. **A speed limit that sees the shape of the spectral measure.** The exact class of populations whose autocorrelation *modulus* is smallest at half period is not known to us, and it is not convex: $(10, 4, 1)/15$ and $(1, 4, 10)/15$ both have their smallest modulus only at $\theta = \pi$, but their average $(11, 8, 11)/30$ vanishes where $\cos\theta = -4/11$. The auxiliary class defined by the real part, $\mathrm{Re}\, F_p(e^{i\theta}) \geq F_p(-1)$, is linear in the populations, hence a convex cone for finitely many levels, which a Fejér–Riesz factorization should describe; for the modulus, Fejér–Riesz applies to the trigonometric polynomial $|F_p(e^{i\theta})|^2 - F_p(-1)^2$, which is quadratic in $p$.

5. **Fejér's four times monotone theorem** [7] in Lean, and the positive trigonometric sums beyond Fejér–Jackson (Vietoris) — the toolkit is in place.

## Questions to take with you

A paper built from questions should end with the ones it could not answer, one per station of the map. Two of them (on Q1 and Q3) are pedagogical rather than research questions, and we say so.

**On Q1.** Weight 1 (logarithms) needs no delicate limits; weight 2 needed exactly one ($\ln x \cdot \ln(1-x) \to 0$, the only place our formalization had to fight). Is the number of genuinely delicate boundary limits an *invariant of the weight* — and what does it count? (To be a mathematical question it needs a notion of "delicate" independent of the choices of a formalization.)

**On Q2.** We reached Landen's golden values by replacing one two-variable identity (Abel's) with three one-variable ones, because the golden point sits where reflection and Landen orbits intersect. Which specializations of the five-term relation are reachable this way — precisely: what

is the subgroup of relations generated by reflection, Landen and duplication inside the full functional-equation group of $\mathrm{Li}_2$, and is the five-term relation in it?

**On Q3.** On the circle, the even-weight cosine series is a polynomial and the sine series is transcendentally new; at odd weight the roles swap. What does weight parity *know* about the boundary that makes it exchange the tame and the wild half? (The classical answer is the Fourier series of the Bernoulli polynomials; the question is how to make a formal library use it uniformly in the weight.)

**On Q4.** Two positivity proofs in this development end the same way: both endpoints at zero and the critical-point equation at an interior extremum (Fejér–Jackson and the Margolus–Levitin cosine inequality). The positivity of $\mathrm{Cl}_2$ is inherited from Fejér–Jackson by Abel summation here, although classically it also follows from $\mathrm{Cl}_2'(\theta) = -\ln(2\sin(\theta/2))$ by the same rise-then-fall shape. Is there a single theorem behind them — a maximum principle for positive sums — and would *it* be the right thing to formalize next?

**On Q5.** The zeta state defeats both textbook speed limits through its infinite moments, and the fractional-moment limits see it only through its lowest level. Exact moments can sometimes pin the whole distribution: for $E \geq 0$ with $\langle E\rangle = \langle E^2\rangle = \langle E^3\rangle = \frac{4}{5}$ one has $\langle E(E-1)^2\rangle = 0$, so the populations are $(\frac{1}{5}, \frac{4}{5})$ on $\{0, 1\}$ and $|S| \geq \frac{3}{5}$, with no dominant lowest level. What can be certified from finitely many moments known only approximately, or for spectral measures with infinite support, where no finite set of moments pins the support? (Zieliński and Zych's doublet example [11] shows how far the moments can move without changing the orthogonalization time.)

**On Q6.** Complete monotonicity and four times monotonicity put the smallest modulus at half period; three times monotonicity and log-convexity (hence convexity) do not. What is the exact class — not a convex one (Section 9) — and is it, as for Hurwitz's theorem, a condition one can read off the differences of the populations?

# A Reproducing

| | version 1 | version 2 |
|---|---|---|
| repository | `godsil-gutman-lean` | `formalizations` |
| commit | `3c09d67` | `c635ec2` |
| Lean | `v4.30.0-rc2` | `v4.35.0` |
| Mathlib | `701fb6e9` | `f0b4c815` |

```
git clone https://github.com/karlesmarin/godsil-gutman-lean
cd godsil-gutman-lean && git checkout --detach 3c09d67
lake exe cache get && lake build Dilog QSL
for script in figures/dilog_fig*.py; do python "$script"; done

git clone https://github.com/karlesmarin/formalizations
cd formalizations && git checkout --detach c635ec2
lake exe cache get && lake build && sh scripts/check_axioms.sh
```

The figure scripts (in `godsil-gutman-lean`, Python with `mpmath` and `matplotlib`) assert their mathematical content against 20–40-digit numerics before drawing; `check_axioms.sh` checks that

every main result depends only on the three standard axioms. The Isabelle sessions build with `isabelle build -d <session folder> <session>` (Isabelle2026). The numbers of Section 7 (the threshold $s^*$, the Zieliński–Zych bounds, the exact examples) come from the scripts in the supplementary material.

# B Inventory of the formalizations

## Lean 4

| File (version 1, `godsil-gutman-lean`) | Contents | Lines |
|---|---|---|
| `Dilog/Basic.lean` | $\mathrm{Li}_2$: series, derivative, reflection, Landen, duplication, $\mathrm{Li}_2(\frac{1}{2})$, golden ladder, Rogers L, Lee–Yang | 581 |
| `Dilog/Clausen.lean` | $\mathrm{Cl}_2$, Catalan's $G$, Bernoulli parabola, $\mathrm{Cl}_2 \geq \sin\theta/2$, $\mathrm{Cl}_2(2\pi-\theta) = -\mathrm{Cl}_2(\theta)$, Theorem 4 | 332 |
| `Dilog/FejerJackson.lean` | Fejér–Jackson inequality | 189 |
| `QSL/Basic.lean` | Margolus–Levitin (+ the cosine inequality) | 238 |
| `QSL/MandelstamTamm.lean` | Mandelstam–Tamm, $L^1$ form | 134 |
| | | 1 474 |

| File (version 2, `formalizations`) | Contents | Lines |
|---|---|---|
| `.../Dilogarithm/CircleFloor.lean` | $\lvert\mathrm{Li}_2\rvert \geq \pi^2/12$ on the circle; $\lvert S\rvert \geq \frac{1}{2}$ | 239 |
| `.../Polylog/HausdorffFloor.lean` | Theorem 5, moment series | 469 |
| `.../SpeedLimit/ZetaStateFloor.lean` | $\eta = (1-2^{1-s})\zeta$; Corollary 1 | 157 |
| `.../SpeedLimit/DominantPopulation.lean` | $\lvert S\rvert \geq 2p_j - 1$ | 84 |
| `.../SpeedLimit/ZielinskiZych.lean` | ZZ inequality and bound; Proposition 1 | 507 |
| `.../Polylog/AntipodalCounterexamples.lean` | the exact examples of Section 7 | 363 |
| `.../Polynomial/EnestromKakeya.lean` | Eneström–Kakeya, Hurwitz's boundary case, series | 350 |
| `.../SpeedLimit/MandelstamTamm.lean` | also $\tau \geq 1/\Delta E$ | 170 |

Table 1: The Lean 4 development. All theorems `sorry`-free; axioms: `propext`, `Classical.choice`, `Quot.sound` throughout.

## Isabelle/HOL

| Session | Contents | Lines |
|---|---|---|
| `Hausdorff_Floor` | per-term bound, uniform gap, floor of moment series | 254 |
| `Polylog_Floor` | Gamma-density moments; Theorem 5; $\eta = (1-2^{1-s})\zeta$; Corollary 1 | 399 |
| `Antipodal_Examples` | $\lvert S\rvert \geq 2p_j - 1$; the exact examples of Section 7 | 539 |
| `Zielinski_Zych` | ZZ inequality and bound ($0 < \alpha \leq 1$); Proposition 1 | 696 |
| | | 1 888 |

Table 2: The Isabelle/HOL development (Isabelle2026, HOL-Analysis; folder `Isabelle/` of [28]): independent proofs of results of the Lean development, with no `sorry` and no `quick_and_dirty`.

# C What changed in version 2

*Corrections.* (1) Priority: the polylogarithm of integer order with its duplication formula was in Isabelle's AFP before version 1 [24]; version 1 said the dilogarithm was absent from every proof

assistant. (2) Theorem 4 follows in one line from $p_1 > \frac{1}{2}$; version 1 presented it as the opposite of what one would expect and said that on this state "the speed limits of quantum mechanics predict nothing at all", which holds only for the two textbook limits. (3) The reason the clock never ticks is the dominant level, not the heavy spectral tail. (4) The floor $|S| \geq \frac{1}{2}$ follows from Fejér (1937), is stated for $Li_2$ by Boyer and Parry, and is now a Lean theorem; version 1 had it numerically and without a citation. (5) Remark 1 of version 1 is Hurwitz's 1913 theorem; version 1 said it had not found it. (6) Version 1 said that three positivity proofs of the development share one shape, one of them "$Cl_2$ itself via the integral of $-\ln(2\sin(t/2))$"; in the Lean, $Cl_2 > 0$ comes from Fejér–Jackson by Abel summation. (7) Version 1 said that every textbook reaches the golden values through Abel's five-term relation and called the route without it a discovery; the values are Landen's, older than Abel's relation. (8) Version 1 said the state "never gets even halfway to distinguishability": $|S| \geq \frac{1}{2}$ means a survival probability of at least $\frac{1}{4}$, while the trace distance can reach $\frac{\sqrt{3}}{2}$. (9) Version 1 spoke of an infinite energy variance and of the textbook bounds being "vacuously satisfied"; with an infinite mean the variance is not defined, and the bounds simply do not apply. (10) Version 1 called the speed-limit statements "fully real"; only Margolus–Levitin is. (11) The figure command of version 1 ran only the first script. *Additions.* Section 7; Proposition 1; the Lean files of the second table.

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